

Nate Woodward

LC T P Spring symposium: Theoretical Physics and AI

Aims of Our Data

1

2



wants:



DON'T wants:

✓ Training data for small
reasoning models

✓ Variable difficulty

✓ Broad topic coverage

✓ Pedagogical topics & difficulty

~~Benchmark for frontier model~~

~~Research level tasks or topics~~

 Tool use

Aims of Our Datasets

Wants:

- ✓ Training data for small reasoning models
- ✓ Variable difficulty
- ✓ Broad topic coverage
- ✓ Pedagogical topics & difficulty

DON'T Wants:

- ✗ Benchmark for frontier model
- ✗ Research level tasks or topics
- ✗ Tool use

Datasets & Problem Seeds

Synthetic

Domain Topic

ADVANCED UNDERGRADUATE	GRADUATE
• Foundations and Scalar Fields • Path Integrals and Quantization • Symmetries and Scattering	• Fermions and Spinor Structure • Gauge Fields and Symmetry • Renormalization and Corrections
ADVANCED GRADUATE	POST GRADUATE
• Renormalization Group and Scaling • Non-Abelian Gauge Theories • Finite Temperature/Vacuum Effects	• Amplitude and On-Shell Methods • Loop and Regularization Techniques • Effective Field Theories

Operational Difficulty

Easy Medium Hard